Fiscal Year:	FY 2006	**Task Last Updated:**	FY 05/30/2006
PI Name:	Shapiro, Bruce Ph.D.		
Project Title:	Discovery and Definition of the Microgravity Transcriptome with Application to Muscle Atrophy		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Zip Code:	91109	**Congressional District:**	29
Comments:			
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Wold, Barbara		
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	To develop and apply microarray data discovery tools, emphasizing informatics and computational modeling, to the biological microgrivity response, thereby analyzing the data and mining from it new specific hypotheses. We will build a database of transcriptome measurements relevant to microgravity, and apply to this data a growing suite of analytical tools, and engage in developing and testing more refined secondary hypothesis about specific pathways that are targets for countermeasures. We will refine and apply existing in-house and commercial computational algorithms such as novel clusting algorithms designed to analyze time-course data and cell simulation programs to generate a modeling framework for signaling pathways and target genes they affect in microgravity. 1. Obtain an objective empirical view of gene expression changes due to muscle atrophy and muscle regeneration, sufficient to enable identification of a large fraction of all genes subject to substantial changes in RNA level during these processes. Make data and primary		

Task Description:	clustering analyses widely available to other NASA researchers via the Web. 2. Create statistical models of the correlations between expression profiles and other tissue sample variables such as microgravity simulation, sex, age, cell type composition, and treatment or stress history. Exploit these statistical models to generate testable hypotheses about the genes and the regulatory relationships involved in the processes of muscle atrophy and regeneration. Build computational hypothesis-formulation methods that can be refined and reused as more data becomes available. Combine with expert user hypothesis building from annotated gene sets. 3. Translate hypothesized sets of regulatory relationships into predictive causal pathway models (circuit models) for gene expression profiles under various conditions (represented by conditions of circuit input variables), using and extending current genetic regulatory network circuit inference techniques. Build a hypothesis-refining and hypothesis–pruning methodology, which can be continually applied as more data becomes available. 4. Based on results from 1-3, design future spaceflight experiments suitable for extracting maximal information about microgravity muscle atrophy and possible interventions, using the above techniques. Alterations in gravity impact cells in many ways, including cell proliferation, chromosomal aberations, gene expression, and processes of reproductive cell formation. At the systemic level, microgravity affects virtually every system in the body. Known detremental changes include bone mineral density losses that vary across specific bone sites with greater loss in the weight-bearing bones. Marked atrophy of postural muscles as well as reduced muscle strength and tasks prevent astronauts from performing sustained work in space and cause severe side-effects upon return to Earth. These effects increase with time in flight and it is not known how long they continue to develop. This project will help to understand the molecular and cellular causes of muscle atrophy and identify potential targets for counteremasure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Muscle unloading samples for rats (hindlimb suspension experiments) were provided by JSC; these samples were subsequently reduced to RNA and stored under refrigeration for experimental analysis. Skeletal muscle samples of soleus, gastrocnemius and tibialis anterior from rats on one week hindlimb suspension protocol were processed for total RNA extraction, along with samples of left cardiac ventricle. Total RNA yield per milligram of tissue mass was significantly reduced for soleus and gastrocnemius compared to ambulatory control rats, confirming previously published results. Total RNA samples were reverse transcribed into double stranded cDNA containing a T7 promoter site, from which biotin labeled cRNA was produced. cRNA samples were analyzed for consistent yield and length distribution by electropherogram, prior to hybridization to Affymetrix Rat Expression set 230 version 2.0 chips. Each chip assays 30,248 expressed sequences and variants using 11 probe pairs per sequence. The average portion of chip capacity reliably detected in the eight hybridizations performed was 46% ± 4%. Data analysis was performed using CompClust and a Cistemic Pipeline. CompClust is a Python package for exploratory data analysis and is based on pyMLX and IPLOT, and includes tools to explore and quantify relationships between data features, clustering results, and data annotations (such as manually-assigned clusters, gene names, and gene ontology (GO) functional groups). The Cistemic pipeline is used to identify and characterize cis-acting regulatory sequences. Mussa is a C++ software package that identifies sequences conserved across multiple genomes. It is unique in that it performs an N-way transitive analysis, and that it has an graphical user interface to interactively explore the results of detailed conserved sequence analyses. Two genes in particular, Muscle Atrophy F-box (MAFbx) and Muscle RING Finger-1 (MuRF-1), are ubiquitin ligases that are required for muscle atrophy and are used as molecular markers of atrophy in many studies searching for mediators of the atrophy response in skeletal muscle. By identifying and functionally characterizing genomic cis-regulatory elements that regulate the MAFbx and MuRF-1 genes, we sought to identify other genes that may share these regulatory sequences using genome-wide searches for consensus/refined binding sites. This information will aid us in the identification of upstream signaling inputs to these critical mediators of the atrophy response, which future countermeasures may target. Our approach involved characterizing the gene expression profiles for each of the skeletal muscles tested, then used Cistematic to identify and refine candidate cis-regulatory elements within conserved regions of genes with similar expression, followed by a genome-wide search for and characterization of other similarly co-regulated genes. Mussa was used to perform a 4-way analysis of MuRF-1 neighborhood to identify regions conserved in human, mouse, dog and chicken. Cistematic then identified candidate cis-regulatory elements within conserved regions of MAFbx, MuRF-1, MuRF-2 in human, mouse and dog genomes.
Bibliography Type:	Description: (Last Updated: 09/19/2006)
Articles in Peer-reviewed Journals	Hart CE, Scharenbroich L, Bornstein BJ, Trout D, King B, Mjolsness E, Wold BJ. "A mathematical and computational framework for quantitative comparison and integration of large-scale gene expression data. " Nucleic Acids Res. 2005 May 10;33(8):2580-94. PMID: 15886390 , May-2005
Articles in Peer-reviewed Journals	Roden JC, King BW, Trout D, Mortazavi A, Wold BJ, Hart CE. "Mining gene expression data by interpreting principal components." BMC Bioinformatics. 2006 Apr 7;7(1):194 [Epub ahead of print] PMID: 16600052 , Apr-2006
Papers from Meeting Proceedings	Roden J, Hernandez G, Williams B, Kuntz S, Mortazevi A, King B, Trout D, Wold B, Shapiro B. "A genome-wide study of cis-regulation within the microgravity-induced muscle atrophy transcriptome. " Pacific Symposium on Biocomputing, Jan 2006. Pacific Symposium on Biocomputing, Jan 2006. Poster presentation. , Jan-2006